MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6550A 1/

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: Beam power pentode

Outline - - - 14-16 (EIA)

Envelope - - - T-14

Base - - - B7-99

Cathode - - - Coated unipotential

Base connections:

Pin No. - - -	1	2	3	4	5	7	8
Element - - -	Base Shell/NC	h	a	g2	g1	h	k, g3

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec2	Ehk	Pp	Pg2	Ec1	Ik	Rg1	TE	Alt
Unit:	V	V dc	V dc	V	W	W	V dc	mA dc	Meg	°C	ft
Maximum - Pentode connection:	6.9	660	440	-300 +200	46	6.6	-300	210	0.05 (Note 2)	250	Note 1
Maximum - Triode connection:	6.9	500	500	-300 +200	52	---	-300	210	0.05 (Note 2)	---	---
Minimum:	5.7	---	---	---	---	---	---	---	----	---	---
TEST CONDITIONS:	6.3	450	450	---	---	---	-48	---	----	---	---

GENERAL:

Qualification - Not required

1/ Direct replacement for tube type 6550

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Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 1	3					
1266	Total grid current	4	Ec2 = 300 V dc; Ec1 = 0; Rk = 200 ohms	Ic1	---	-5	µA dc
1256	Electrode current (1) (anode)			Ib	45	105	mA dc
1341	Power output (1)		Eb = Ec2 = 250 V dc; Ec1 = -14 V dc; Esig = 9.8 V ac; Ef = 5.5 V; Rp = 1,500 ohms	Po	8.4	---	W
1246	Audio frequency noise		Esig = 280 mV ac; Eb = Ec2 = 250 V dc; Ec1 = -14 V dc; Rp = 1,500 ohms	EB	---	17	vu
1256	Electrode current (1) (screen)		Eb = 150 V dc; Ec2 = 170 V dc; Ec1 = -10 V dc	Ic2	---	9.5	mA dc
1201	Short and discontinuity detection			---	---	---	---
	Quality conformance inspection, part 2						
1336	Heater-cathode leakage	5		Ihk	---	100	µA dc
1256	Electrode current (2) (screen)	5		Ic2	---	8.5	mA dc
1341	Power output (2)		Eb = Ec2 = 250 V dc; Ec1 = -14 V dc; Esig = 9.8 V ac; Rp = 1,500 ohms	Po	10	---	W
1256	Electrode current (2) (anode)	5	Ec2 = 300 V dc; Ec1 = -60 V dc	Ib	---	1.0	mA dc
1266	Grid emission	6	Ef = 7.5 V; Eb = Ec2 = 250 V dc; Ec1 = -70 V dc; Rg1 = 0.05 Meg	Ic1	---	-5.0	µA dc
1211	Insulation of electrodes	7		---	---	---	---
1301	Heater current			If	1.38	1.62	A
1101	Secureness of base and base insert			---	---	---	---
1111	Base pin solder depth			---	---	---	---
1105	Permanence of marking			---	---	---	---

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Method	Requirement or test	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
	Quality conformance inspection, part 3						
1506	Heater-cycling life		Ef = 8.0 V; 1 minute "on", 4 minutes "off"; Ehk = +200 V dc; Eb = Ec1 = Ec2 = 0	---	---	---	---
1501	Intermittent life		Group A; Eb = 400 V dc; Ec2 = 300 V dc; Ehk = +200 V dc; Rk = 180 ohms; Rg1 = 0.05 Meg	---	---	---	---
---	Intermittent life-test end point (500 hours):						
1341	Power output (2)		Eb = Ec2 = 250 V dc; Ec1 = -14 V dc; Esig = 9.8 V ac; Rp = 1,500 ohms	Po	9.6	---	W
1131	Metallic base sleeve quality	8		---	---	---	---
1216	Base material insulating quality	8	Zone 5, minimum	---	---	---	---

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. The grid resistance rating given is for fixed - bias operation; for cathode - bias operation, Rg1 shall be 0.25 Meg maximum.
3. The AQL (per cent defective) for each test listed under QCI, part 1, shall be 0.65, inspection level II.
4. This test shall be performed at the conclusion of the holding period.
5. The AQL (per cent defective) shall be 4.0, inspection level II.
6. Prior to this test, tubes shall be preheated a minimum of 5 minutes at the conditions indicated below. The 3-minute test shall not be permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid emission test.

Ef	Eb	Ec2	Ec1	Rk	Rg1
V	V dc	V dc	V dc	Ohms	Meg
7.5	350	350	-19	100	0.05

7. The AQL (per cent defective) shall be 4.0, inspection level S3.

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NOTES: - Continued

8. This test shall be performed during the initial production and once each succeeding 12-calendar month period in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes having an acceptance number of zero, and a second sample of three tubes having a combined acceptance number of one. In the event of failure, the test shall be made as a part of quality conformance inspection, part 2, AQL 6.5, inspection level S3. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

9. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - MI
Navy - AS, CG, MC, OS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5960-3330)

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